



CAP599: Next Gen Computing Project-I

Title: Effects of social media on mental health

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1. Nature of Project/Research Work:

This project is a research-based analytical system that studies the impact of social media usage on mental health. It is a web-based and AI-driven predictive framework that integrates sentiment analysis and fuzzy logic modeling to evaluate emotional well-being and behavioral risks associated with excessive social media usage.

The project leverages data analytics, machine learning concepts, and rule-based inference systems to analyze user behavior such as screen time, self-comparison habits, distraction levels, and sleep disturbances. The objective is to develop a predictive model that assesses potential mental health risks and provides interpretable insights into emotional patterns influenced by digital media usage.

2. Proposed Topic/Idea Name:

Title: Effects of social media on mental health

Effects of social media on mental health - This project focuses on analyzing how social media platforms influence individuals' psychological well-being, emotional stability, sleep patterns, and productivity. By combining behavioral analysis with sentiment classification, the system aims to measure and predict mental health risk levels in a structured and interpretable manner.

3. Source of the Topic:

The project is self-initiated and inspired by the increasing psychological challenges observed among youth due to excessive social media consumption. With rising concerns about anxiety, depression, sleep disorders, self-esteem issues, and digital addiction, this research aims to provide a data-driven and technology-based solution to understand and measure these impacts scientifically.

4. Brief Description of the Project:

Social media platforms such as Instagram, WhatsApp, YouTube, Facebook, and Twitter have become an integral part of daily life. While these platforms provide connectivity and entertainment, excessive usage has been linked to emotional

distress, reduced concentration, sleep disturbances, and self-comparison tendencies.

This project proposes an analytical framework that collects user data through structured surveys and applies data preprocessing, sentiment classification, and fuzzy inference techniques to assess mental health risks. The system evaluates behavioral parameters such as:

- Daily screen time
- Frequency of aimless scrolling
- Self-comparison with others
- Sleep quality
- Emotional responses to social media usage

The project uses sentiment analysis to classify textual responses into positive, neutral, or negative emotional states. A fuzzy rule-based inference system is then applied to predict mental health risk levels, particularly focusing on sleep disturbance risk.

By combining behavioral data and emotional analysis, the system provides an interpretable prediction score that reflects the potential psychological impact of social media usage

Key Features

- 1) **Data Collection and Preprocessing:** The system collects user data through structured surveys and performs data cleaning and preprocessing to ensure accurate analysis.
- 2) **Sentiment Analysis:** Textual responses are analyzed and classified into Positive, Neutral, or Negative categories. These sentiments are converted into numerical scores for further modeling.
- 3) **Behavioral Feature Analysis:** Key factors such as screen time, self-comparison habits, distraction levels, and sleep quality are extracted to identify patterns related to mental stress.
- 4) **Fuzzy Logic-Based Prediction Model:** A fuzzy inference system evaluates combined behavioral and emotional inputs to categorize mental health risk levels (Low, Moderate, High).
- 5) **Sleep Risk Assessment:** The system predicts the likelihood of sleep disturbances caused by excessive social media usage.

- 6) **Demographic Insights:** Comparative analysis is performed across age and gender groups to understand variations in emotional and behavioral impact.

5. Project/ Outcome:

The outcome of this project will be the development of a structured analytical model designed to evaluate the psychological impact of social media usage. The system will generate a predictive mental health risk score, analyze emotional sentiment distribution, and provide behavioral insights related to digital habits. By integrating these components, the project offers data-driven awareness regarding the potential risks of social media addiction and excessive usage. Furthermore, it promotes healthier digital practices by presenting measurable insights into how online engagement influences emotional and psychological well-being. The study also establishes a foundation for future advancements in AI-based mental health monitoring systems and real-time intervention tools.

Conclusion

The project “**Effects of social media on mental health**” aims to bridge the gap between psychological research and technological solutions. By integrating sentiment analysis and fuzzy logic modeling, the system provides a structured and interpretable approach to understanding the emotional and behavioral consequences of excessive social media usage.

With increasing digital dependence in modern society, this project highlights the importance of responsible social media usage and provides a data-driven framework for identifying potential mental health risks. Future enhancements may include advanced deep learning models, chatbot-based emotional support systems, and real-time monitoring dashboards to further strengthen mental health awareness and digital well-being initiatives.